UNICC AI Safety Lab Capstone Part (C): UX Design & NYU AI Sandbox Deployment

Applied Technical Project
Final Report

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Spring 2026

A paper submitted in partial fulfillment of the requirements for the degree of
Master of Science
at the
Division of Programs in Business
School of Professional Studies
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Abstract

This applied project delivered Part (C) of the UNICC AI Safety Lab Capstone by producing a human-centered user experience (UX) and a reproducible deployment package for an AI safety evaluation prototype in the NYU AI Sandbox. The work addressed a recurring gap in prototype delivery: stakeholders may be unable to evaluate, interpret, or reliably demonstrate results when workflows are fragmented and deployments are not repeatable. Human-centered design practices (user journey mapping, wire-framing, and iterative prototype reviews) were combined with reproducibility practices (dependency pinning, documented setup steps, clean-install verification, and an operator run-book). The resulting solution defined an end-to-end safety testing workflow and produced deployment and documentation artifacts that reduce demo failure risk under sandbox constraints. The work improved stakeholder clarity and reduced operational risk by making execution more consistent and outputs easier to understand.

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Acknowledgements

I sincerely thank Dr. Andres Fortino for serving as my Applied Technical Project Sponsor and United Nations International Computing Centre (UNICC) for providing the opportunity to conduct this Applied Technical Project. I also thank Prof. Israel Moskowitz for the Applied Technical Project course and mentoring, and I appreciate the feedback from classmates and faculty that supported milestone improvements.

Declaration

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1. Introduction

This Applied Technical Project was conducted for the United Nations International Computing Centre (UNICC) as part of the NYU SPS Applied Project course. UNICC provides shared information and communications technology (ICT) services and operational support for organizations across the United Nations system. In this environment, prototypes are evaluated not only for technical capability but also for reliability, transparency, documentation quality, and the ability to demonstrate consistent results to multiple stakeholders. Because UN-related projects often involve cross-functional review and governance expectations, a solution that works only on a developer's machine is typically not considered acceptable. A deliverable must be understandable, repeatable, and auditable enough to be trusted by non-developer stakeholders.

This project focused on Part (C) of the UNICC AI Safety Lab Capstone: building the human interface (UX) and deployment reproducibility foundation for an AI safety evaluation prototype in the NYU AI Sandbox. The project sponsor was Dr. Andres Fortino, who provided expectations and acceptance criteria for the Part (C) deliverables. The approved proposal timeline was February 1, 2026 through May 1, 2026, and the project reached practical completion by April 14, 2026 with time reserved for verification and final documentation polish. Importantly, Part (C) does not redesign the core agent logic itself (handled by Parts A/B). Instead, Part (C) ensures that stakeholders can run the

workflow end-to-end, interpret outputs correctly, and reproduce execution under sandbox constraints.

Because the final evaluation depends on a demonstration, the project prioritized “demo readiness” as a measurable outcome rather than an informal goal. Demo readiness includes: (1) a clear end-to-end workflow (inputs → configuration → run → results), (2) stable execution in the sandbox environment, (3) predictable outputs and error handling, and (4) supporting documentation such as a user guide, operator runbook, and demo checklist. These elements directly reduce the probability of failure during stakeholder review and improve the clarity of what results mean and how they were produced.

2. Problem Description / Opportunity

AI safety evaluation prototypes often show promising technical behavior but fail to translate that behavior into a usable stakeholder evaluation experience. A common pattern in early prototypes is a fragmented workflow: evaluation steps are distributed across scripts, notebooks, configuration files, and manual setup instructions. This fragmentation increases the cognitive load on evaluators because they must remember steps, understand dependencies, and interpret results without consistent structure.

Even if the prototype is technically correct, inconsistent workflow design creates confusion and makes it difficult for stakeholders to trust or compare results across runs.

In addition, the NYU AI Sandbox introduces execution constraints that create "operational failure modes" independent of model quality. Common sandbox problems include dependency mismatches, environment drift, restricted permissions, package installation limitations, and compute/time constraints. These issues are especially damaging in demo situations because they occur at the worst time—during a live run—often forcing a presenter to troubleshoot instead of communicating project outcomes. From a stakeholder perspective, repeated execution failures weaken confidence not only in the deployment but also in the validity of the evaluation results. Therefore, the project's problem is not just a usability issue, but an acceptance and trust issue.

The opportunity for Part (C) is to convert a technically oriented prototype into an evaluation-ready deliverable by standardizing the evaluator workflow and controlling operational risk. A clear UX flow enables stakeholders to understand what inputs were used, what configuration was applied, what was tested, and what evidence supports the results. Reproducible deployment practices enable the same workflow to run consistently in the sandbox, minimizing the "works on my machine" effect. Together, these improvements reduce demonstration risk, lower stakeholder confusion, and make the prototype easier to extend by future teams without rebuilding the interface and deployment process from scratch.

3. Importance of the Project to the Organization

For UNICC and UN-related environments, stakeholder trust is a primary success condition for AI initiatives. Trust is built through transparency (what was tested and why), repeatability (results can be reproduced), and clear communication (outputs are interpretable). AI safety evaluation is particularly sensitive because stakeholders are looking for evidence that the system has been tested responsibly and that findings are not artifacts of one-off runs. If the evaluation workflow is unclear or cannot be reproduced, the organization's ability to rely on evaluation results is reduced, and adoption decisions may be delayed or rejected.

Part (C) directly supports organizational needs by improving the operational readiness of the capstone prototype. A consistent interface and repeatable deployment reduce the time and effort required to conduct evaluations, prepare demonstrations, and onboard new users. In practical terms, this means fewer "setup and troubleshooting" cycles, faster execution of review sessions, and clearer outputs during decision-making discussions. It also reduces risk during high-stakes demos by providing stable procedures, contingency approaches, and documentation that can be followed by someone other than the original developer.

In addition, Part (C) creates reusable assets that can be carried forward into future iterations of the AI Safety Lab work. The user journey, interface structure, runbook, and demo checklist establish a foundation that future teams can extend as the underlying

agent capabilities evolve. Instead of rebuilding UX and deployment processes repeatedly, the organization gains a structured approach that supports continuous improvement while preserving stakeholder-facing consistency.

4. Alternate Solutions Evaluated

Several alternatives were evaluated for delivering Part (C) under the constraints of a sandbox-based prototype environment. The first alternative was a notebook-only workflow in which evaluation is conducted through scripts and notebooks with manual documentation of steps. This approach has low development overhead and is common in early prototypes, but it is difficult for non-technical stakeholders to use and is highly sensitive to environment issues. It also increases demo risk because success depends on manual step execution and troubleshooting under time pressure.

A second alternative was a minimal front-end prototype with mocked outputs. This approach improves presentation clarity because the flow can be demonstrated without depending on backend integration stability. However, it does not satisfy operational readiness objectives: stakeholders cannot run the workflow end-to-end, and the deployment reproducibility problem remains unsolved. As a result, a mock-only approach may be acceptable for early concept demonstrations but does not meet acceptance expectations for a capstone deliverable meant to be executed in a managed environment.

The selected approach was an integrated UX workflow combined with reproducible deployment practices aligned to Parts (A/B). This approach offered the best balance of usability and reliability. The UX flow makes evaluation steps explicit and reduces cognitive load for stakeholders, while deployment controls (dependency pinning, clean-install thinking, runbooks, and checklists) reduce the probability of sandbox-related failures. Although this approach requires more engineering effort than the alternatives, it best supports stakeholder acceptance because it improves clarity, repeatability, and demonstration reliability.

5. Solution Evaluation Criteria

- Usability: evaluators can complete the core workflow without developer assistance.
- Clarity: outputs show what was tested, key metrics/flags, and concise summaries with optional drill-down details.
- Repeatability: deployment and execution can be reproduced in the NYU AI Sandbox using pinned dependencies and documented steps.
- Integration readiness: the interface aligns with Parts (A/B) inputs/outputs and tolerates controlled schema evolution.
- Demo reliability: a demo checklist and fallback plan reduce probability of demo failure.

6. Selection Rationale

The integrated UX and reproducible deployment approach was selected because demonstration reliability and stakeholder clarity were primary acceptance drivers. The approach also supports extensibility by standardizing configuration inputs and result formatting, enabling future teams to iterate on the underlying agent while maintaining a stable evaluator experience.

7. Approach and Methodology

A milestone-driven approach was used to structure work into initiation/planning, environment discovery, UX requirements and design, implementation and integration, testing and stabilization, and deployment/documentation readiness. Cross-team dependency risk with Parts (A/B) was managed by defining expected input/output schemas and by using stubs/mocks to maintain UX progress when upstream changes occurred.

UX methodology followed human-centered AI design practices: user journey mapping, wire-framing, prototype iteration, and walkthrough reviews to ensure stakeholders could complete the five core tasks and interpret outputs. Deployment methodology emphasized reproducibility: dependency pinning, documented setup steps, clean-install verification, and creation of an operator run-book and demo checklist to support repeatable execution in the NYU AI Sandbox.

Methodology details (UX): The UX workflow was designed as a guided, task-oriented sequence rather than a collection of screens. Each step aligns with a user question an evaluator would naturally ask: (1) What input am I evaluating? (2) What safety checks am I running and with what parameters? (3) What is happening right now? (4) What did the system find? (5) How do I inspect evidence and explain findings to others? Design decisions prioritized clarity over density. Defaults were used to reduce configuration burden, and advanced options were intentionally de-emphasized to keep the core workflow simple for first-time reviewers.

Methodology details (interpretability): To support reliable interpretation, results were structured into summaries and evidence. Summaries provide high-level flags and aggregated metrics that allow quick triage. Evidence views show the underlying examples—prompts, responses, and the reason for a flag—so stakeholders can validate that a metric corresponds to observable behavior. This design reduces the risk of over-trusting a single score and supports governance review because the decision rationale can be inspected directly. Where appropriate, the interface communicates uncertainty and avoids absolute language when results reflect probabilistic signals.

Methodology details (testing): Testing followed a scenario-based approach that mirrors evaluation use. Positive-path tests confirmed that all five core tasks can be completed end-to-end. Negative-path tests validated resilience to missing inputs, invalid parameters, and runtime exceptions likely to occur in a sandbox environment. Each failure mode was mapped to a user-facing message that explains what happened and

what to do next. This approach reduced ambiguity and improved demo reliability because it prevented silent failures and minimized last-minute troubleshooting. Methodology details (deployment hardening): Deployment steps were treated as a product deliverable. The project created a repeatable setup process that includes dependency pinning, deterministic configuration defaults, and a clean-install verification. A clean-install verification means reinstalling from a fresh state and executing a minimal smoke test to confirm that the environment supports the workflow. This was combined with a rollback approach: if a sandbox refresh introduces incompatibilities, the deployment can revert to a last known-good dependency set and re-validate using the same smoke test procedure. These steps reduce risk from environment drift and increase confidence that demonstrations will be repeatable.

8. Project Objectives and Metrics for Project Evaluation

Project Goal: Deliver Part (C) by building a usable UX (human interface) and a reproducible NYU AI Sandbox deployment package for the AI Safety Agent prototype, including user guide, operator run-book, and demo checklist (Part C only; excludes building the MoE agent itself).

1. Define UX requirements and acceptance criteria by Week 3; sponsor/team approval.
2. Deliver a clickable prototype or UI skeleton by Week 5, supported by at least 3 walkthrough reviews and documented feedback.

3. Implement the competition-ready UX and integrate with Parts (A/B) so 5 core user tasks run end-to-end by Week 8.
4. Deploy the integrated solution to NYU AI Sandbox by Week 10 with reproducible setup/run steps and validated redeploy (clean install test).
5. Finalize documentation and demo readiness by end of semester: user guide + operator run-book + demo checklist; gather ≥5 reviewer feedback with average ≥4/5 usability score.

9. Risk Analysis

Risk management in this project focused on operational and integration risks typical of sandbox-based AI prototypes. The highest operational risk was sandbox environment drift, where a previously working configuration fails due to changes in dependencies, system packages, or runtime constraints. This risk was addressed by controlling dependencies, documenting installation steps, and validating that the workflow can be executed consistently under sandbox conditions. A second operational risk was performance and timeout risk: even correct code can fail in a constrained environment if runtime exceeds limits or memory usage spikes. To mitigate this, the project emphasized demo-safe scenarios and maintained the ability to reduce workload if performance degraded.

Integration risk with Parts (A/B) was also significant. Part (C) consumes upstream outputs; if output format changes late, interface binding can break and require rework. This risk was managed by aligning on an “integration contract” concept: expected schema fields, stable structure, and sample outputs used for UI development. Stubs and mocks were used strategically so that UX development could continue even when upstream outputs were still evolving. This reduced schedule exposure and protected demo readiness.

Finally, usability and interpretation risk was treated as a real acceptance risk. Even if the system runs, stakeholders may reject it if results are confusing or if the workflow does not support review and discussion. Mitigation strategies included walkthrough reviews, layered result presentation (summary first, drill-down second), visible system status, and actionable error messages. These controls reduced the probability that stakeholders would misinterpret outputs or lose confidence during evaluation sessions.

10. Issues Encountered

While no project-stopping issues occurred, several practical issues were encountered and managed during delivery. The first category involved environment and dependency consistency. Sandbox constraints can cause small differences in package behavior or availability, leading to runtime exceptions that do not appear locally. These issues were addressed by pinning dependencies where possible, documenting environment

assumptions explicitly, and using a repeatable setup process rather than ad-hoc installation commands.

The second category involved integration alignment with Parts (A/B). Because Part (C) depends on upstream output structure, minor changes in field names, nesting structure, or output formats required updates in how results were displayed. This was managed through coordination checkpoints and by keeping stubs/mocks available so UX iteration could continue without being blocked. When integration changes occurred, the project prioritized preserving the five core user tasks and ensured that the user journey remained stable even if specific output details evolved.

The third category involved usability refinements discovered through walkthrough reviews. In early versions, certain labels, status messages, or result summaries were technically accurate but not intuitive to stakeholders. This was corrected by simplifying terminology, making system status more explicit (what step is running and what comes next), and presenting results in a layered way that supports quick understanding before deep inspection. These usability-driven fixes reduced confusion and improved demo confidence.

11. Detailed Project Chronology and Critique

- Initiation and planning: project scope, acceptance expectations, and timeline were confirmed; a baseline plan (WBS/Gantt) was established.

- Environment discovery: NYU AI Sandbox constraints and operational assumptions were documented to inform UX and packaging decisions.
- Requirements and acceptance criteria: core workflow tasks and success metrics were defined to guide design and verification.
- UX design: user flows and wireframes were developed and refined; a prototype/UI skeleton enabled walkthrough reviews and rapid iteration.
- Integration alignment: expected input/output schema and sample outputs were coordinated with Parts (A/B); stubs/mocks reduced dependency delays.
- Implementation and stabilization: core workflow was implemented and tested end-to-end for the five core tasks; error-handling and result presentation were refined.
- Deployment and documentation: reproducible setup/run procedures were prepared; user guide, operator runbook, and demo checklist were produced; dry-run demos were conducted.

Critique: earlier formalization of versioned schema contracts and automated contract tests would further reduce late-stage integration uncertainty. Future iterations should add automated sandbox smoke tests as part of the deployment workflow to detect environment drift immediately after sandbox refreshes.

Expanded chronology (Week-by-week narrative): The project timeline was organized to reduce late-stage demo risk by front-loading constraint discovery and stakeholder alignment. In Weeks 1–2, planning focused on translating the proposal into an executable schedule and confirming the definition of “done” for Part (C). This included

clarifying that Part (C) would deliver an evaluator-facing workflow and deployment reproducibility artifacts—not core agent modeling changes. During this phase, the work breakdown structure (WBS) was created to make dependencies explicit and to identify the critical path leading into the final demo window.

In Weeks 2–3, environment discovery and feasibility checks were prioritized. The NYU AI Sandbox introduces unique constraints compared to local development: package availability may be limited, compute resources may be capped, and network/permission policies may restrict access to external services. The project treated these constraints as product requirements. The team documented what could be assumed (e.g., allowed runtime, allowable dependency installation method) and what must be validated repeatedly (e.g., whether a sandbox refresh changes preinstalled libraries). This work reduced the risk of "environment drift," where a previously working build fails later because the environment changed.

In Weeks 3–4, requirements were refined into an acceptance-criteria format. Instead of treating UX as subjective, requirements were expressed in terms of observable evaluator tasks. The five core tasks—selecting inputs, configuring a test, executing the run, reviewing results, and recovering from common errors—were used as the acceptance baseline. This framing allowed walkthrough reviews to be conducted as task completion exercises rather than general design opinions, making feedback easier to prioritize and implement.

In Weeks 4–5, user flows and wireframes were developed to determine the information hierarchy needed for safety evaluation. Safety testing outputs can be complex; therefore, the interface was designed to present results in layers. The first layer provides an executive summary (what was tested, top-level flags/metrics, and overall status). The second layer supports drill-down into individual cases, including the prompt, model response, and the specific reason a case was flagged. This layered design supports both non-technical stakeholder review and technical debugging without overwhelming either audience.

In Weeks 5–6, a prototype/UI skeleton was used to validate interaction choices. The prototype emphasized clarity of labels, reduction of cognitive load, and explicit system status. For example, during execution, the UI communicates what stage the run is in and whether the system is waiting on model output, computing metrics, or formatting results. This supports the human–AI interaction guideline of making system status visible and reduces user confusion during longer runs. Error messaging was written to be actionable (what happened, why it matters, and what to do next) rather than generic (e.g., “failed”).

In Weeks 6–7, integration planning focused on controlling dependency risk with Parts (A/B). Because Part (C) depends on upstream interfaces, the project established a practical “integration contract” concept: a shared understanding of input fields, output schemas, and sample outputs used for UI binding. Stubs and mocks were kept in place so that the UI could be exercised even if upstream outputs changed. This approach reduced

schedule risk by allowing the interface to mature independently while still remaining compatible with the evolving agent components.

In Weeks 7–9, implementation and stabilization focused on end-to-end task completion. The primary test was not individual functions, but whether an evaluator could complete the five core tasks without manual intervention. Testing also included negative paths: missing required inputs, invalid configuration values, and runtime failures triggered by sandbox limitations. These tests were used to improve resilience and to ensure the final demo experience would be predictable. Defects were triaged using a simple rule: issues that block core tasks or obscure result interpretation were prioritized over cosmetic improvements.

In Weeks 9–10, deployment packaging and reproducibility checks were finalized. This included locking dependency versions, documenting installation steps, and conducting "clean install" verification—rebuilding from a fresh environment state to confirm reproducibility. The operator runbook was written as a procedural guide that can be followed by someone other than the developer, and the demo checklist was created to reduce presentation risk. A fallback plan was also established (demo-safe test set and precomputed outputs) to mitigate performance/timeouts under sandbox compute constraints.

Critique (what could be improved): The project's largest risk driver was cross-component change. While stubs/mocks reduced the cost of waiting, future iterations would benefit from automated contract tests that fail fast when upstream schemas change.

Additionally, maintaining a versioned library of “golden” evaluation runs—inputs and expected outputs—would enable regression testing of both UX presentation and agent behavior. Finally, earlier access to definitive sandbox constraints (including refresh schedules and package whitelists) would reduce uncertainty and improve the efficiency of the deployment hardening phase.

12. Lessons Learned

1. Operational readiness matters as much as technical capability.

Early in the project, it became clear that stakeholders judge a prototype by whether it can be run smoothly and repeatedly, not only by whether the underlying model or logic is correct. A solution that works only in a developer environment creates high demo risk and reduces stakeholder confidence. The key lesson is to treat “demo-readiness” as a real deliverable: clear steps, predictable runtime, and understandable outputs. In future work, I would define “ready” using checklists and dry-run success criteria from the beginning rather than waiting until the end.

2. Sandbox constraints should be treated as requirements, not surprises.

The NYU AI Sandbox environment introduces practical constraints such as package limitations, runtime restrictions, and environment drift. These are not minor details—they shape what can be delivered and how stable the final demo will be. The lesson learned is to validate the environment early and repeatedly, and to design the workflow so it remains usable even under constraints. Next time, I would schedule recurring

sandbox smoke tests and maintain a small “known-good” demo configuration that can run reliably.

3. Cross-team dependencies require explicit interface contracts.

Part (C) depends on Parts (A/B) outputs. Without clear agreements on input/output schema and example outputs, interface changes can create rework late in the timeline. The lesson is that integration risk is best managed by writing down a simple “contract” early: required fields, output structure, and sample outputs for UI binding. In future projects, I would add schema versioning and lightweight contract tests (even simple checks) so breaking changes are detected immediately.

4. Stubs/mocks protect progress and reduce schedule risk.

Using stubs/mocks allowed the UX workflow to be implemented and reviewed even when upstream components were still changing. This reduced idle time and kept the project moving forward. The lesson is that decoupling strategies are not shortcuts—they are professional project controls that maintain momentum and reduce uncertainty. In future work, I would formalize this approach: define stub outputs as part of the plan and keep them updated in parallel with integration.

5. Walkthrough reviews are the fastest way to improve usability and clarity.

Iterative walkthroughs exposed gaps that would not be obvious from development alone—for example, unclear labels, confusing status feedback, and results that were technically correct but not stakeholder-friendly. The lesson is that usability issues are easier to fix early than late, and a small number of structured reviews can produce major

quality gains. Next time, I would schedule walkthroughs at fixed milestones and capture feedback in a consistent format (issue → impact → fix → re-test).

6. Good error handling reduces both confusion and demo risk.

In demo-oriented projects, failures are inevitable (timeouts, missing inputs, environment limitations). What matters is whether the system fails gracefully with actionable messages. The lesson learned is to design error states as part of the core product: explain what happened, why it matters, and what the user can do next. In future work, I would build a short "failure mode library" early and test those cases intentionally so the interface remains stable under stress.

7. Reproducibility requires verification, not only documentation.

Writing setup instructions is helpful, but it does not guarantee reproducibility. The project reinforced that the real test is a clean-install verification: reinstall from scratch and confirm the workflow runs. The lesson is to treat reproducibility as something you validate repeatedly, especially when environments can change. In future projects, I would run clean-install checks on a schedule (e.g., weekly) and before every milestone demo.

13. Conclusions and Summary

This applied project successfully delivered Part (C) of the UNICC AI Safety Lab Capstone by producing a stakeholder-oriented UX workflow and a reproducible deployment foundation for the AI safety evaluation prototype in the NYU AI Sandbox. The core

contribution is an end-to-end evaluation experience that is easier to run, easier to interpret, and more reliable to demonstrate than an ad-hoc notebook/script workflow. The project emphasized clarity of user journey (inputs → configuration → run → results), transparent communication of system status, and structured result presentation so stakeholders can quickly understand what was tested and what findings were produced.

In addition, the project reduced operational risk through reproducibility practices: dependency control, clearly documented setup steps, verification through clean-install thinking, and operational artifacts such as a user guide, operator runbook, and demo checklist. These deliverables strengthen stakeholder confidence by making the evaluation workflow repeatable and auditable in practice, not only in theory. Overall, the project improved the readiness of the capstone prototype for demonstration and evaluation, and it provides a foundation that future teams can extend as the underlying agent capabilities evolve.

14. Limitations, Recommendations and Scope for Future Work

Limitations:

- Scope limited to Part (C).

  This project focused on UX and deployment reproducibility and did not modify the underlying MoE agent logic (Parts A/B). Therefore, the evaluation depth and accuracy are constrained by the outputs and interfaces provided by the upstream components.

Part (C) improves usability and operational readiness, but it does not change the core model behavior itself.

- Sandbox constraints limit integration options.

  NYU AI Sandbox restrictions may limit access to certain packages, network resources, persistent storage, or monitoring tools that would be standard in production environments. As a result, some production-grade features (persistent run history dashboards, external observability, automated monitoring alerts) are not included in this delivery.

- Evaluation coverage is bounded by available test scenarios.

  Because the deliverable is demo-focused, evaluation scenarios were prioritized for clarity and reliability rather than maximum breadth. This means the prototype may not fully represent performance under a wide variety of real-world edge cases until more scenarios and testing time are added.

Recommendations and future work:

- Implement schema versioning + automated contract checks.

  To reduce late-stage integration rework, future teams should version the Parts (A/B) output schema and implement simple automated checks to detect breaking changes. Even lightweight tests that validate required fields and output structure can prevent unexpected UI failures close to demos.

- Add exportable governance-ready reporting.

  A high-value enhancement is to allow exporting run results and metadata

(configuration, scenario coverage, flagged items, summary metrics) into formats such as CSV/PDF. This would support governance workflows and enable stakeholders to share findings more easily beyond the interface.

- Expand failure taxonomy and annotation workflows.

  Future iterations can improve interpretability by adding structured labels for failure modes (e.g., hallucination, sensitive content, refusal failure, policy violations) and allowing reviewers to annotate results. This would make the prototype more useful for systematic red-teaming and mitigation planning.

- Improve performance profiling and reliability under constraints.

  To reduce timeout risks, future work should include performance profiling, adaptive batching, and configurable timeouts. A “demo mode” with a smaller test set and cached/precomputed outputs should remain available as a contingency for live demonstrations.

- Add longitudinal comparison and run history.

  A strong future feature is the ability to compare runs over time (before/after changes), store run history, and visualize trends. This would turn the prototype from a single-run demonstration tool into a practical evaluation system that supports continuous improvement.

15. Literature Survey

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1. Introduction

This literature survey synthesizes practice-based research relevant to delivering Part (C) of the UNICC AI Safety Lab Capstone: designing a usable human interface (UX) and creating a reproducible deployment package for an AI safety evaluation prototype in the NYU AI Sandbox. Because this project is evaluated through demonstrations and operational readiness, the survey focuses on best practices that can be translated into implementation decisions. The review emphasizes three areas: (1) AI risk and safety evaluation, (2) human-centered AI interface design and interpretability, and (3) reproducible deployment and MLOps practices. Together, these themes support a core goal of Part (C): making the system easy to run, easy to understand, and reliable to demonstrate.

2. Survey Scope and Source Selection

Sources were selected to cover: (a) AI governance and risk-management frameworks relevant to public-sector contexts, (b) evaluation approaches for large language models (LLMs) and AI systems, (c) design guidance for human-facing AI workflows, and (d) reproducibility and production-oriented ML practices that reduce “works on my machine” failures. The survey draws on peer-reviewed research, widely adopted

frameworks (e.g., NIST), and reputable technical documentation that is commonly used in operational settings.

3. AI Risk Management and Governance Foundations

Public-sector and UN-related environments typically require AI systems to be trustworthy, auditable, and aligned with organizational risk posture. The NIST AI Risk Management Framework (AI RMF 1.0) presents a lifecycle approach to identifying, assessing, and managing AI risks through governance, mapping, measurement, and management activities. For a capstone prototype, the key implication is not building comprehensive governance tooling, but ensuring that evaluation outputs, settings, and limitations are presented transparently so stakeholders can reason about risk and oversight expectations.

A consistent theme in responsible AI is the need to document model and data characteristics so stakeholders understand intended use, limitations, and failure modes. Model Cards propose a standardized reporting artifact describing model details, performance considerations, and key cautions. Datasheets for Datasets similarly argue for standardized documentation describing dataset motivation, composition, collection process, and recommended uses. While Part (C) does not build the underlying models or datasets, these documentation practices directly inform what the UX should surface: what was evaluated, under what assumptions, and what caveats apply.

4. AI Safety Evaluation: Benchmarks, Red Teaming, and Multi-Metric Assessment

AI safety evaluation increasingly relies on multi-metric measurement beyond accuracy (e.g., robustness, bias, toxicity, calibration, efficiency). HELM (Holistic Evaluation of Language Models) exemplifies a standardized approach that combines scenario coverage with multi-metric reporting to improve transparency of LLM behavior. For Part (C), HELM reinforces two practical design principles: evaluation should be organized around scenarios, and reporting formats must be consistent so results are comparable and understandable to stakeholders.

Red teaming is another important practice in AI safety evaluation. Anthropic's work describes systematic probing to identify and reduce harmful outputs, along with lessons learned from iterative mitigation. From a UX standpoint, red teaming implies that evaluation tools should support: (1) organizing prompts/scenarios, (2) tagging or categorizing discovered failure modes, and (3) making results reviewable by non-developer stakeholders. In other words, Part (C) should not only run tests; it should help users interpret and communicate outcomes.

In engineering workflows, evaluation frameworks help teams define test cases, run evaluations repeatedly, and track performance changes. OpenAI Evals illustrates how eval registries, data, and results can be structured for repeatability. Even if Part (C) does not adopt Evals end-to-end, the framework supports patterns that Part (C) can borrow: standardized test configurations, run metadata capture, and stable result presentation.

5. Human-Centered AI UX: Interaction Design and Interpretability

Human-facing AI systems work best when users develop correct mental models of capability and limitations. Guidelines for Human–AI Interaction propose 18 empirically validated design guidelines spanning the interaction lifecycle. For Part (C), the most actionable implications include: making system status visible, providing meaningful explanations, supporting user control (e.g., changing evaluation parameters), and designing for graceful failure with clear error messages and recovery paths.

Interpretability and explanation are often essential for trust, especially when outputs influence decisions. Doshi-Velez and Kim emphasize that interpretability must be evaluated relative to user context and that explanations should be fit-for-purpose rather than one-size-fits-all. A practical Part (C) translation is “layered explanation”: provide a high-level summary for quick understanding, with drill-down views for technical validation.

LIME is a widely cited method for local explanations and demonstrates how explanation artifacts can help users judge trust and identify errors. Even if LIME itself is not used, the broader UX lesson applies: explanations must clarify what is being explained, avoid overstating certainty, and connect outputs to input conditions. For an AI safety testing interface, this means clearly showing prompts, model responses, scores/flags, and why an item was categorized as risky.

6. Deployment Reproducibility and MLOps Practices

In applied AI systems, engineering risk often dominates: hidden dependencies, integration complexity, and environment mismatch can derail delivery. Sculley et al.

describe “hidden technical debt” in ML systems, where data dependencies and glue code accumulate and make systems fragile. For Part (C), the message is direct: a working interface is not sufficient if the deployment cannot be reproduced reliably in the sandbox.

Reproducibility is supported through disciplined practices such as dependency pinning, clear documentation, and routine re-runs in clean environments. Pineau et al. summarize reproducibility initiatives and provide concrete engineering practices that improve repeatability. For Part (C), this implies treating the sandbox deployment as a reproducibility target: lock dependency versions, document setup steps precisely, and validate installation regularly to prevent environment drift.

Production ML platforms provide patterns for repeatable pipelines and validated artifacts. TFX describes a production-scale ML platform emphasizing pipeline components, validation, and artifact consistency. Even if Part (C) does not use TFX directly, the architecture reinforces practical best practices: scripted setup, deterministic defaults, and early checks that fail fast when prerequisites are missing—especially important under sandbox constraints.

7. Synthesis: Best Practices Mapped to Part (C) Deliverables

7.1 UX deliverables

Across the literature, usability and trust depend on clear workflows and correct user mental models. For Part (C), this supports a UI structure that includes: (1) clear task entry points (e.g., “Run safety test”), (2) explicit configuration panels with sensible

defaults, (3) transparent run status and logs, and (4) results views that summarize findings while allowing drill-down into flagged examples and failure modes. Documentation practices such as Model Cards and Datasheets also inform what the UX should capture and display—intended use, limitations, scenario coverage, and evaluation settings—so stakeholders interpret results responsibly.

7.2 Evaluation and reporting deliverables

Evaluation research emphasizes multi-metric, scenario-based reporting to avoid over-reliance on a single score. Red teaming research emphasizes structured discovery and categorization of harmful behaviors. For Part (C), this implies presenting results as a small dashboard of metrics and categories, with the ability to inspect individual examples and annotate failure modes. Standardized result schemas (inspired by evaluation frameworks) reduce ambiguity and improve repeatability across runs.

7.3 Deployment and reproducibility deliverables

MLOps and reproducibility research highlights that systems fail operationally when dependencies and configuration are not controlled. Part (C) should therefore deliver a deployment package with pinned dependencies, scripted setup, and a “clean install” verification procedure. Because sandbox environments can reset or update, proactive validation reduces demo risk and supports stakeholder confidence.

8. Gaps in the Literature and Practical Implications

A key gap is that many evaluation and governance frameworks define what should be measured and documented, but provide limited concrete guidance on UX patterns for

communicating results to non-technical stakeholders. Human-centered AI guidelines help bridge this gap, but they must be adapted to the specific workflows of safety testing. Therefore, Part (C) should validate UI concepts through rapid prototypes and walkthroughs to ensure stakeholders can complete tasks and correctly interpret outputs. Another gap is the difference between research evaluation and constrained sandbox deployment. Production platforms assume full DevOps access, but sandbox deployments often restrict networking, packages, and permissions. This increases the importance of clear runbooks, minimal dependencies, and fallback demo modes (e.g., precomputed outputs) to mitigate timeouts or environment drift.

9. Conclusion

The literature converges on a consistent message: successful AI safety evaluation is not only a modeling problem—it is also operational and human-centered. To support trustworthy evaluation, Part (C) must deliver a clear workflow, interpretable results, and reproducible deployment. Guidance from governance frameworks, evaluation research, human-centered AI design, and reproducibility/MLOps practices provides concrete best practices that reduce project risk and improve demo readiness.

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16. Situational Analysis

Applied Project Situational Analysis

UNICC AI Safety Lab Capstone Part (C): UX Design & NYU AI Sandbox Deployment

Student: Qing Wang

Organization: United Nations International Computing Centre (UNICC)

Date: March 26, 2026

1. Executive Summary

This situational analysis summarizes the external and internal environment influencing delivery of UNICC AI Safety Lab Capstone Part (C), which focuses on UX design and reproducible deployment of an AI safety evaluation prototype in the NYU AI Sandbox. The analysis reviews: (1) industry context, (2) competitive landscape, and (3) organizational culture. Porter's Five Forces and a SWOT analysis are applied to identify forces that may affect project success and to translate those forces into actionable delivery strategies.

2. Project Context and Relationship to the Organization

Project scope (Part C) in brief:

- Define the end-to-end user journey for running safety tests (inputs → configuration → run → results).
- Design and deliver a usable interface (prototype and/or UI skeleton) aligned with NYU AI Sandbox constraints.

- Deploy the integrated solution to the NYU AI Sandbox with reproducible steps (packaging, setup, run, rollback).
- Deliver operational documentation: user guide, operator runbook, and demo checklist.

Relationship to other parts of the work/program:

- Depends on Parts (A) and (B) for core MoE agent logic, test harness, and output format; Part C consumes these interfaces.
- Supports UNICC/AI Safety Lab evaluation goals by making the prototype operable, demo-ready, and easier for stakeholders to understand.
- Aligns with broader UN technology governance needs (responsible AI, auditability, and repeatability), even under constrained sandbox conditions.

3. Industry Analysis

Industry context: UN shared services / public-sector IT services; AI safety and responsible AI enablement. UNICC functions as a technology and digital services provider for UN entities. As public-sector organizations explore AI adoption, expectations increase for transparency, risk management, and evidence that systems can be evaluated and controlled. For Part (C), this environment elevates the importance of clear evaluation workflows and reproducible runs because stakeholders must be able to understand and trust how findings were produced.

Macro forces influencing the project:

- Regulatory & governance expectations: Responsible AI, explainability, audit trails, and risk documentation raise the bar for how evaluation results are presented and reproduced.

- Technology trends: Evaluation tools and deployment practices change quickly; reproducible deployment reduces demo and handoff risk amid change.
- Stakeholder perception & trust: Adoption depends on confidence; clear UX and repeatable runs improve trust and decision-making.
- Operational constraints: Sandbox restrictions on packages, network access, and compute shape what can be delivered and demonstrated.

4. Competitive Landscape

In a capstone/prototype context, “competition” is best interpreted as substitute solutions that satisfy the same need: enabling AI safety evaluation, monitoring, or governance reporting. Relevant competitor/substitute categories include:

- AI safety evaluation toolkits & benchmarks: Open-source frameworks can be adopted quickly; differentiation comes from workflow quality and usability.
- Model monitoring / governance platforms: Commercial platforms provide dashboards and audit workflows; they substitute for custom UX if operational readiness is prioritized.

- Internal evaluation scripts and ad-hoc processes: Low-cost substitutes (notebooks/scripts) exist but are fragile for demos and hard to standardize.
- Other academic or lab prototypes: Alternatives may offer similar functionality; a reliable, clear demo experience becomes a differentiator.

Implication for Part C: To outperform substitutes, Part (C) must emphasize simplicity of workflow, interpretability of outputs, and reproducible sandbox deployment. These elements turn a prototype into something stakeholders can reliably evaluate.

5. Organizational Culture and Internal Environment

UNICC and UN-related environments commonly prioritize reliability, security, stakeholder alignment, and documentation. Decision-making can involve multiple stakeholders, and success depends on predictable execution and artifact quality. For Part (C), this means that clear communication, explicit acceptance criteria, and operational documentation (runbooks and checklists) are not “extra”—they are core to perceived deliverable quality.

Culture-related factors relevant to Part (C):

- Stakeholder-driven expectations: Frequent feedback cycles and clear criteria reduce late surprises.

- Documentation and transparency: User guides and clear outputs support handoff and evaluation.
- Risk sensitivity: Demonstrating controls, reproducibility, and auditability supports confidence.
- Cross-team dependencies: Integration with Parts (A/B) requires coordination and change control to avoid rework.

6. Porter's Five Forces Analysis

Porter's Five Forces is used to analyze the environment around AI safety evaluation solutions and why Part (C) should prioritize UX clarity and reproducible deployment:

- Competitive rivalry: High — Many evaluation approaches exist; demos are judged by usability, clarity, and reliability, not only novelty.
- Threat of new entrants: Medium — New tools appear quickly; a modular, documented deployment reduces switching costs.
- Threat of substitutes: High — Notebooks or governance platforms can replace a custom UX; Part (C) must add value through workflow and repeatability.
- Bargaining power of buyers (stakeholders): High — Stakeholders can reject confusing or non-repeatable solutions; UX and documentation strongly influence acceptance.

- Bargaining power of suppliers: Medium — Dependencies include sandbox constraints and upstream Parts (A/B) interfaces; change control and version pinning reduce risk.

7. SWOT Analysis (Project and Organization Context)

This SWOT focuses on factors most relevant to Part (C) delivery:

Strengths

- Clear Part (C) scope: UX + deployment + documentation.
- Ability to iterate quickly through prototypes and walkthrough reviews.
- Stakeholder value is easy to demonstrate through usability and repeatability.

Weaknesses

- Strong dependency on Parts (A/B) interface stability.
- Sandbox constraints may limit tooling and integration options.
- Time-limited usability testing compared to production environments.

Opportunities

- Differentiate by making outputs interpretable and demos reliable.
- Create reusable deployment/runbook assets for future teams.

- Use acceptance criteria to align stakeholders early and reduce rework.

Threats

- Environment drift breaking reproducible deployment.
- Late upstream changes causing rework.
- Demo timeouts/performance issues under constrained compute.

Strategies derived from SWOT

- Lock an integration contract (schema + sample outputs) with Parts (A/B); use stubs/mocks to decouple UI progress.
- Pin dependencies and test clean installs regularly to prevent environment drift; maintain rollback steps.
- Define a demo-safe test set and a performance budget; provide a backup demo mode (precomputed results) if needed.
- Run structured walkthroughs to validate the five core user tasks; simplify flows when confusion is observed.

8. Implications for Project Strategy

Based on the forces above, Part (C) should prioritize:

- Reproducibility first: version pinning, documented setup, repeatable sandbox runs.

- Stakeholder clarity: outputs designed for explainability (what was tested, what failed, why).
- Dependency management: early agreement on input/output schema and change control with Parts (A/B).
- Demo readiness: define demo scenarios early, test repeatedly, and maintain a fallback plan.

9. Conclusion

This situational analysis indicates that Part (C) success depends most on usability, stakeholder trust, and operational repeatability under sandbox constraints. Competitive substitutes and high stakeholder expectations mean the project must deliver a clear workflow and stable deployment, supported by strong documentation and a controlled integration contract. These strategies reduce schedule risk and improve readiness for the final demonstration and evaluation.

Appendices

Appendix A. Sponsor Project Acceptance Document (Doc 07 Part 1 and Part 2)

Figure A-1. Doc 07 Part 1 (Sponsor Acceptance – Plan)

Sponsor's Project Acceptance Document – PART 1

This document is the means by which your project sponsor formally agrees that your project has been satisfactorily completed and that it meets the project goal and objectives that were set at the onset of the project. It is therefore important that you describe the goal, objectives, and related metrics in the appropriate section below. The "PLAN" section is to be prepared at the beginning of the project and the "RESULTS" and "ACCEPTANCE" sections after your project has been completed. Your sponsor should provide input and sign where indicated. The signed document will also be a required section in your Project Final Report. This document is a

Project Title: UNICC AI Safety Lab Capstone Part (C): UX Design & NYU AI Sandbox Deployment

Student Name: Qing Wang

Sponsoring Organization: United Nations International Computing Centre (UNICC)

Project Sponsor Name and Title: Dr. Andres Fortino (Project Sponsor)

Project Sponsor Contact Information (email and phone): agf249@nyu.edu

PROJECT PLAN

At project start, show the project goal; the project objectives and related metrics to be used to show successful project completion. Sponsor should sign to indicate agreement.

Project Goal: Deliver Part (C) by building a usable UX (human interface) and a reproducible NYU AI Sandbox deployment package for the AI Safety Agent prototype, including user guide, operator runbook, and demo checklist (Part C only; excludes building the MoE agent itself).

Objective #1: Define UX requirements and acceptance criteria by Week 3

Objective #2: Deliver a clickable prototype or UI skeleton by Week 5, supported by at least 3 walkthrough reviews and documented feedback.

Objective #3: Implement the competition-ready UX and integrate with Parts (A/B) so 5 core user tasks run end-to-end by Week 8.

Objective #4: Deploy the integrated solution to NYU AI Sandbox by Week 10 with reproducible setup/run steps and validated redeploy (clean install test).

Objective #5: Finalize documentation and demo readiness by end of semester: user guide + operator runbook + demo checklist.

ADD as many additional objectives as necessary, at least three, preferably five.

I agree with the above planned project goal, project objectives, and related metrics.

Andres Fortino — **Project Sponsor Signature**

4/24/2026 — **Date:**

Printed Name: **Andres Fortino**

07-Part1 SponsorsProjectAcceptanceDocument V3.3.docx

Figure A-2. Doc 07 Part 2 (Sponsor Acceptance – Results & Acceptance)

Sponsor's Project Acceptance Document -Part 2

PROJECT RESULTS

STUDENT: Include below the project goal and ALL the objectives in your project, which you detailed in Project Plan section (07-Part1). You may add additional objectives, or add space, as needed.

Planned Start Date: 02/01/2026 **Planned End Date: 05/01/2026**
Actual Start Date: 02/02/2026 **Actual End Date: 04/14/2026**

If actuals differ from planned dates, the revised dates (Actual) are accepted by the sponsor if initialed here: _____ **Sponsor Initials** _____

Project Goal Was the project goal achieved as planned? ☑Yes ☐No, Reason missed: _____
If NO, please explain why this is an acceptable deviation. _____ **Sponsor Initials** AF

Project Objective #1: Define UX requirements and acceptance criteria by Week 3.
Did the student's project meet this objective with associated measures and metrics as established at project inception? **Objective#1** ☑has or ☐has not been met. **Sponsor Initials** AF
If not met please explain why this is or is not an acceptable deviation.

Project Objective #2: Deliver a clickable prototype or UI skeleton by Week 5, supported by at least 3 walkthrough reviews and documented feedback.
Did the student's project meet this objective with associated measures and metrics as established at project inception? **Objective#2** ☑has or ☐has not been met. **Sponsor Initials** AF
If not met please explain why this is or is not an acceptable deviation.

Project Objective #3: Implement the competition-ready UX and integrate with Parts (A/B) so 5 core user tasks run end-to-end by Week 8.
Did the student's project meet this objective with associated measures and metrics as established at project inception? **Objective#3** ☑has or ☐has not been met. **Sponsor Initials** AF
If not met please explain why this is or is not an acceptable deviation.

Project Objective #4: Deploy the integrated solution to NYU AI Sandbox by Week 10 with reproducible setup/run steps and validated redeploy (clean install test).
Did the student's project meet this objective with associated measures and metrics as established at project inception? **Objective#3** ☑has or ☐has not been met. **Sponsor Initials** AF
If not met please explain why this is or is not an acceptable deviation.

Project Objective #5: Finalize documentation and demo readiness by end of semester: user guide + operator runbook + demo checklist
Did the student's project meet this objective with associated measures and metrics as established at project inception? **Objective#3** ☑has or ☐has not been met. **Sponsor Initials** AF
If not met please explain why this is or is not an acceptable deviation.

PROJECT ACCEPTANCE

Sponsor: Please show your evaluation of student's work, add comments, sign below

SPONSOR
: Select

Sponsor's evaluation of student's performance on this project:

☐ **A** ☐ **A-** ☐ **B+** ☐ **B** ☐ **B-** ☐ **C+** ☐ **C** ☐ **C-** ☐ **F**

Sponsor's Comments: _____

_____

SPONSOR:
Select One

☑ Project was completed satisfactorily and is hereby accepted.

☐ Project was completed satisfactorily but did not meet all objectives, as shown above. Nevertheless, the Project is accepted.

☐ Project was NOT completed satisfactorily and is NOT accepted.
Please explain why not accepted in Sponsor's Comments section, above.

Andres Fortino	4/24/2026
Project Sponsor Signature	**Date:**
	4/24/2026
Student Signature	**Date:**

All signatures in English. Sign by hand or use e-signature. Do not type or use handwriting font.

Appendix B. Project Plan, Baseline WBS/Gantt

Figure B-1. Baseline WBS/Gantt

WBS–Gantt Chart (Outline View) — Qing Wang

Project: UNICC AI Safety Lab Capstone Part (C) — UX & NYU AI Sandbox Deployment

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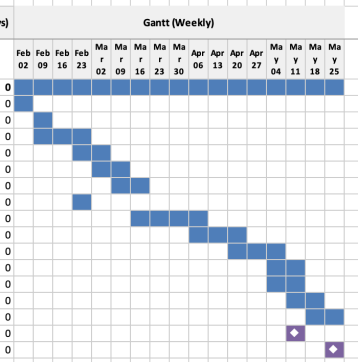
ID	Task Name (WBS)	Duration (days)	Start	Finish	Predecessors	Resources	Baseline Start	Baseline Finish	Baseline Duration	Finish Variance (days)
1	**1 Capstone Part (C) — UX & NYU AI Sandbox Deployment**	**84**	**2/2/26**	**5/28/26**		**QW**	**2/2/26**	**5/28/26**	**84**	**0**
2	1.1 Initiation & planning	5	2/2/26	2/6/26		QW	2/2/26	2/6/26	5	0
3	1.2 Confirm sandbox constraints & access	3	2/9/26	2/11/26	2	QW;Admin	2/9/26	2/11/26	3	0
4	1.3 UX requirements + acceptance criteria	8	2/12/26	2/23/26	3	QW;Team;Sp	2/12/26	2/23/26	8	0
5	1.4 User flows + information architecture	5	2/24/26	3/2/26	4	QW	2/24/26	3/2/26	5	0
6	1.5 Wireframes + review 1	5	3/3/26	3/9/26	5	QW;Team	3/3/26	3/9/26	5	0
7	1.6 Clickable prototype + reviews 2–3	8	3/10/26	3/19/26	6	QW;Team	3/10/26	3/19/26	8	0
8	1.7 Integration contract (API/IO schema)	4	2/24/26	2/27/26	4	QW;Team	2/24/26	2/27/26	4	0
9	1.8 Implement UX (core screens + error states)	15	3/20/26	4/9/26	7,8	QW	3/20/26	4/9/26	15	0
10	1.9 Integrate UX with backend (a/b) + stubs	10	4/10/26	4/23/26	9	QW;Team	4/10/26	4/23/26	10	0
11	1.10 System test (5 core tasks) + fixes	8	4/24/26	5/5/26	10	QW;Team	4/24/26	5/5/26	8	0
12	1.11 Deploy to NYU AI Sandbox (reproducible)	6	5/6/26	5/13/26	11	QW;Admin	5/6/26	5/13/26	6	0
13	1.12 Docs (user guide + runbook + demo checklist)	6	5/6/26	5/13/26	11	QW	5/6/26	5/13/26	6	0
14	1.13 Pilot demo + feedback + refinements	5	5/14/26	5/20/26	12,13	QW;Team;Sp	5/14/26	5/20/26	5	0
15	1.14 Final release package + presentation readiness	5	5/21/26	5/27/26	14	QW;Team	5/21/26	5/27/26	5	0
16	1.15 Milestone: Deployment & docs approved	0	5/14/26	5/14/26	12,13	QW;Sp	5/14/26	5/14/26	0	0
17	1.16 Milestone: Final demo / submission	0	5/28/26	5/28/26	15	QW	5/28/26	5/28/26	0	0



Legend: Task bar; ◆ Milestone

Deviation summary: The project reached practical completion by April 14, 2026, earlier than the planned end date of May 1, 2026. No scope expansion was required to achieve the Part (C) goal and objectives. Remaining time in the original plan was used for verification, documentation polish, and dry-run demonstrations.

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